Windows Server, Active Directory Lab Prep, and Enterprise Infrastructure Foundations

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 2 hours.

Windows Server Foundations

Windows Server is a server operating system developed by Microsoft. It is designed to manage networks, users, applications, databases, and other IT services in businesses and organizations. Windows Server is designed to manage networks and enterprise services, while Windows 11 is designed for everyday personal use. Enterprises use Windows Server because it allows them to centrally manage large numbers of computers, users, security policies, and business services from one platform. Without servers, every computer would need to be managed individually, which becomes impossible at enterprise scale.

So: Windows Server is designed for: centralized management, enterprise infrastructure, authentication, networking, virtualization, file sharing, and remote administration. It is NOT normal desktop usage.

Research EACH: (what it does, why enterprises use it, security importance)

- Active Directory Domain Services (AD DS): Core directory service in Windows Server environments. It stores and manages: users, computers, groups, passwords, permissions, and policies. Enterprises use it to manage thousands of employees from one system, allow single sign-on, organize departments with Organizational Units (OUs), apply Group Policies, and centrally control access to resources.
- DNS Server: converts human-readable names into IP addresses, enterprises use DNS so systems can locate servers, find domain controllers, access websites/services by name, and communicate across networks.
- DHCP Server: automatically assigns network settings to devices: IP address, subnet mask, default gateway, DNS servers. Managing thousands of IP addresses manually would be impossible
- File Server: centrally stores and shares files over the network using SMB, allows centralized storage, collaboration, backup, etc…
- Print Server:
- Hyper-V: Microsoft's virtualization platform, allows one physical server to run multiple VMs. this reduces hardware costs, isolates workloads, improves scalability, creates testing environments, and runs cloud infrastructure
- Remote Desktop Services

Domain Controller (REVIEW): Remember, a Domain Controller is a server running Active Directory Domain Services that manages authentication, authorization, and identity inside a

Windows domain environment. A domain is a centralized network environment where users and computers are managed together. Instead of every PC having separate accounts: one central system controls logins and permissions. Ex:

COMPANY DOMAIN: corp.local

Users: jsmith, abrown, itadmin

Computers: HR-PC01, SALES-LAPTOP03, SERVER01

Employees can log into multiple company computers using the same credentials because the Domain Controller verifies them.

It authenticates users by verifying their credentials against the Active Directory Domain Services database. In modern Windows domains, this is usually done with Kerberos authentication.

Authentication Process:

1. User Enters Credentials
2. Computer uses DNS and Active Directory records to locate a Domain Controller.
3. Computer sends a request to the DC's Key Distribution Center. The request says: "I want to authenticate as this user."
4. DC checks: username exists, password hash matches, account enabled, account not locked out
5. If successful, DC issues a Ticket Granting Ticket. This proves the user is authenticated.

If a DC fails, the impact depends on whether the organization has multiple DCs or only one. If they have only one: users may not be able to log in, new authentication requests fail, Group Policy stops updating, network resources become inaccessible, and servers relying on AD may break. Essentially: enterprise identity system becomes unavailable.

If they have multiple: another DC takes over.

Active Directory Infrastructure Concepts

Again: AD is the centralized system for managing: users, computers, passwords, permissions, groups, policies, and devices inside a Windows domain network.

An Active Directory Forest is the highest-level structure in an AD environment. It contains: one or more domains, shared schema, shared configuration, and trust relationships.

```
FOREST
│
├── sales.company.com
├── hr.company.com
```

├── japan.company.com
└── research.company.com

Each domain is like a department or division. The forest is the entire company structure connecting them together.
Forest Components:

1. Domains
2. Shared Schema: defines what types of objects exist and what attributes they have
    a. Ex: User object:
    b. - username
    c. - email
    d. - password hash
3. Global Catalog: helps users search for objects across all domains
4. Trust Relationships: domains inside the forest automatically trust each other, this allows for cross-domain authentication, shared access, and enterprise-wide logins

An Active Directory Tree is a group of one or more domains inside an Active Directory forest that share: a continuous namespace and automatic trust relationships
company.com
│
├── hr.company.com
├── sales.company.com
└── japan.company.com

Group Policy (REVIEW): Windows feature used in AD DS environments that allows administrators to centrally configure and enforce settings on domain computers and user accounts.
Common Security Settings Enforced Through Group Policy:

1. Password Policies
2. Account Lockout Policies (failed login attempts)
3. Windows Firewall Rules
4. Windows Defender Settings
5. USB / Removable Device Restrictions
6. Software Restrictions
7. Windows Update Policies

There is more but I stopped here.

SRV records is a special type of DNS record used to tell computers where a specific network service is located. An SRV record says: what service exists, which server provides it, and what port it uses. AD relies heavily on SRV records so computers can locate: DCs, Kerberos services, LDAP services, and Global Catalog servers.

IP address exhaustion happens when a network runs out of available IP addresses to assign to devices.

NTFS (New Technology File System): primary file system used by modern Windows OS.

- Read, Write, Execute

Share permissions control access over the network. NTFS permissions control access on the filesystem itself.

Windows Administration + Powershell

Run: **Get-Service.** Then: **Get-Service | Sort-Object Status.** Then: **Get-Service | Where-Object {$_.Status -eq “Running”}**

Some Windows Services critical for enterprise environments:

- AD DS
- DNS Server
- DHCP server
- Kerberos
- SMB
- Remote Desktop Services
- Windows Event Log Service
- Windows Defender Service
- Print Spooler Service
- Hyper-V Services
- Windows Time Service (W32Time): Synchronizes system clocks, Kerberos authentication heavily depends on accurate time
- Group Policy Client Service

Run: **Get-EventLog -LogName Security -Newest 20** … Then replace **Security** with **System**

Centralized logging is the process of collecting logs from many systems and sending them to one central location for: monitoring, searching, analysis, alerting, and investigations.

Run: **whoami /groups.** Then: **net localgroup administrators.** Then: **net user**

Run: **Get-ChildItem C:\Users**. Then: **Get-ChildItem C:\Windows\Temp**

VMWare Lab Planning

VMnet is VMware’s virtual networking system. It creates virtual networks that virtual machines use to communicate: with each other, with the host machine, with the internet, or with the physical network. An isolated lab network is a private, separated network environment used for testing, learning, labs, malware analysis, AD practice, and penetration testing. It will not affect real/home networks, enterprise systems, or the internet.